

Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

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Abstract

Quoted stablecoin price gaps are not trades. We distinguish between *optical* dislocations, where a quoted basis exceeds a threshold, and *executable* ones, where net profit survives VWAP slippage, taker fees, and settlement latency after routing through a live order book. Building Stablecoin StressBench on the March 2023 USDC/SVB event, we find that 34.3% of minutes show a > 10 bps optical dislocation while only 2.88% are executable, yielding a $12\times$ price-to-execution gap (block bootstrap 95% CI: $8\times$ – $24\times$), with no calm-trained model closing it and the hindsight oracle exceeding +225 bps net on the executable arbitrage task. Cross-mechanism meta-labeling trained on Terra/LUNA achieves +82.5 bps on the SVB test split (50.8% of oracle returns), demonstrating that order-book execution microstructure transfers across structurally distinct stress mechanisms. A three-way RL diagnostic then establishes that the unconditioned PPO-GRU degenerates to NoTrade at 2.3% positive density, the conditioned variant at 17% density trades 919 times but earns only -29.2 bps, and that only meta-labeling’s explicit positive-label supervision reaches +82.5 bps, showing that positive-label density is necessary but not sufficient for positive returns.

Keywords

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

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1 Introduction

During stablecoin stress events, peg fragmentation creates cross-venue price gaps that appear to be arbitrage opportunities. In practice, VWAP slippage, taker fees, and CEX-to-CEX settlement latency jointly eliminate most apparent profit. We call dislocations where the quoted basis exceeds a threshold *optical*, and those where net profit survives execution costs *executable*.

Prior machine-learning work on stablecoin predictability uses price-based labels to define “profitable” windows, which conflates the optical and executable layers and systematically overstates how predictable actual trading opportunities are. Stablecoin StressBench is designed to close this gap. It provides per-minute execution-aware labels that separate three dislocation layers: an *optical* dislocation where the quoted basis exceeds a threshold, an *executable* one where net profit is positive after VWAP execution and transaction costs at a specified notional, and a *predictable* one where a model correctly identifies the window ex ante. The distance between these layers is the paper’s core measurement.

We build the benchmark on the March 2023 USDC/SVB stress event, where Silicon Valley Bank’s collapse triggered a brief USDC depeg. Using Binance L2 order-book data¹ and a full VWAP book-walking model, we find that 34.3% of 1-minute windows show a > 10 bps price basis but only 2.88% remain net-profitable at \$10K notional: a $12\times$ price-to-execution gap. The headline result is robust to buy-leg proxy error: a 20% depth haircut moves the executable rate from 2.88% to 2.40% and widens the ratio from $12\times$ to $14\times$, with the oracle remaining above +130 bps. We then construct a model ladder spanning rule-based thresholds, statistical baselines, gradient-boosted classifiers, sequence models (GRU), and meta-labeling filters, evaluating each model’s ability to identify the 2.88% of executable windows ex ante. Every model trained on calm-control data loses money on executable arbitrage tasks, and the hindsight oracle gap exceeds 260 bps. Cross-mechanism meta-labeling, trained on a different stress event (Terra/LUNA 2022), is the only approach that achieves positive net returns (+82.5 bps, 50.8% oracle capture).

Contributions.

- (1) An execution-aware benchmark with per-minute labels, documented order-book depth provenance, VWAP book-walking, fees, and a hindsight oracle ceiling, accompanied by an 18-event stress taxonomy serving as data-tier governance and an oracle gap decomposition into false-positive, timing, and sizing components. (§4, §3.3; Tables 4, 3, 5)
- (2) Cross-mechanism transfer evidence: meta-labeling trained on Terra/LUNA achieves +82.5 bps (50.8% oracle capture) on the SVB test split, which to our knowledge is the first demonstration that order-book execution microstructure transfers across structurally distinct stress mechanisms, on the single cross-mechanism pair with route-complete test data. (§6.5, Table 8)
- (3) An RL diagnostic establishing that positive-label density is necessary but not sufficient: a conditioned PPO-GRU at 17% density earns -29.2 bps versus meta-labeling’s +82.5 bps at identical density, isolating explicit positive-label supervision as the additional requirement. (§6.8, Table 10)

2 Related Work

The literature most relevant to our work spans stablecoin stress events, limits to arbitrage, and financial machine learning.

Prior work on stablecoin stress provides the empirical backdrop for our benchmark. Uhlig [1] analyses the Terra/LUNA collapse, whose algorithmic redemption spiral provides a structurally distinct trigger from the SVB bank shock, making it a clean out-of-distribution test for cross-mechanism transfer. Griffin and Shams [2] document that quoted prices can reflect non-fundamental demand, motivating our separation of optical from executable dislocations.

¹The BTC-USDC buy leg uses kline-proxy depth for the SVB window; all other legs use real L2. The 20% depth haircut sensitivity in §6.1 bounds this approximation.

Hernandez Cruz et al. [21] study USDC-USDT DEX liquidity during the SVB collapse and confirm the importance of execution-layer analysis.

The limits-to-arbitrage literature establishes the theoretical foundations that our labels operationalise. Shleifer and Vishny [4] and Gromb and Vayanos [5] show that capital constraints and execution frictions bound arbitrage even when price gaps are observed. Makarov and Schoar [7] demonstrate that cross-venue crypto price gaps are not executable profits once latency and book depth are accounted for, directly motivating our 12 $\times$ price-to-execution gap. Hautsch et al. [3] quantify settlement latency as a first-order cost, which we implement as the fixed $\delta=5$ bps penalty. Cont and Kukanov [6] establish VWAP execution price as the correct profitability object, which our book-walking model in §4.2 implements directly. Gogol et al. [18] introduce a Maximal Arbitrage Value between a USDC-ETH AMM pool and Binance, and our oracle extends this to route-level CEX simulation and model-evaluation benchmarking.

The financial machine learning literature informs both our methodology and our positioning. López de Prado [8] introduces meta-labeling as a technique for filtering a primary signal with a secondary model that learns the conditions under which the primary signal is reliable. We extend this architecture to the cross-mechanism setting in §6.5, training on Terra/LUNA and evaluating on SVB, a transfer that to our knowledge has not been attempted before. Cintra and Holloway [9] apply BOCPD [15] to detect the Terra depeg from Curve AMM pool reserves and demonstrate cross-event transfer via AMM depletion signals. However, their signals differ fundamentally from CEX cross-quote basis, which is noisier and mean-reverting, and no oracle evaluation is reported. On reinforcement learning, Moallemi and Wang [16] and Ning et al. [17] establish the finite-horizon MDP framing for optimal execution that we adopt in §5.2. Mohl et al. [22] demonstrate the feasibility of GPU-scale multi-agent RL on production limit-order-book data, training IPPO on approximately 400 million orders for a joint execution and market-making task. Olby et al. [23] construct a reactive Simudyne simulator that decomposes execution costs into market impact and execution risk, benchmarking RL timing policies against an Almgren–Chriss efficient frontier. Both works confirm that RL can learn effective execution policies when the environment is reactive. Our benchmark contributes the complementary finding that in non-reactive historical replay with structurally rare positive labels, positive-label density and explicit supervision are necessary preconditions. No prior stablecoin-stress benchmark provides one-minute executable labels, route-level depth and fee modelling, a hindsight oracle ceiling, and cross-mechanism transfer evaluation jointly. StressBench is the first to do so.

3 Benchmark Design

3.1 Data Sources and Instruments

The final dataset contains 56,134 1-minute bars across six windows (Table 2). Price features come from Binance Vision (klines, aggTrades) and Coinbase REST candles. Net-profit labels use Binance BTC-USDC and BTC-USDT depth plus Coinbase BTC-USD as the reference leg.

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows	
Buy BTC-USDC	kline-proxy [†]	real L2 (futures)	[†] BTC-USDC
Cross USDC-USDT	real L2 (Mar 12–20)	real L2 (futures)	
Sell BTC-USDT	real L2 (futures)	real L2 (futures)	
Ref BTC-USD	candle-proxy	candle-proxy	

perpetual listed Jan 2024; all 2022 windows kline-proxy.

Table 2: Dataset splits (56,134 1-min bars).

Split	Event	Min.	Pos.
Train	Calm control ($\times 3$)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

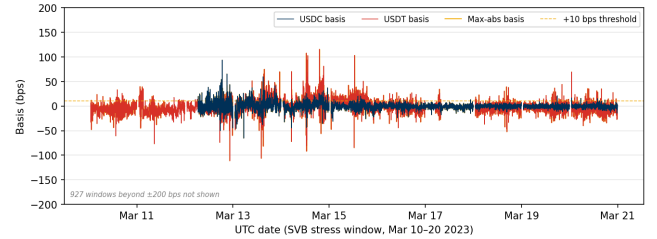


Figure 1: Cross-quote basis (bps) during the SVB test window (Mar 10–20 2023). USDC basis (navy): 12.4% of minutes > 10 bps; USDT basis (red): 27.4%; max-abs (gold): 34.3%. The price-to-execution gap is 12 $\times$.

Every depth record is tagged with a provenance field (Table 1). The sell leg (BTC-USDT) uses real L2 futures bookDepth for all 2023–2024 windows. The cross leg (USDC-USDT) uses real L2 from 2023-03-12 onward. The buy leg (BTC-USDC) uses kline-proxy for the SVB window (BTC-USDC perpetual did not exist on Binance until 2024) and real L2 for 2024 windows. To bound this proxy error, we apply a 20% depth haircut to the kline-proxy buy leg. The executable rate falls from 2.88% to 2.40%, the ratio widens from 12 $\times$ to 14 $\times$, and the oracle remains above +130 bps, confirming that the headline result is robust to buy-leg proxy error.

3.2 Event-Based Split Design

Splits are defined by event identity (Figure 1). No model trained on the train split has seen stress dynamics, and threshold calibration on validation uses an independent stress event. Labels are constructed by aligning each bar’s outcome to the observation window ending at $t+h$, with explicit checks confirming no future information leaks into the input features.

3.3 Historical Stress Taxonomy: Data-Tier Governance

The 18-event taxonomy (2020–2023) serves as a benchmark governance artefact specifying which events carry sufficient data quality for execution-grade claims and which are restricted to price magnitude estimates or mechanism classification only. It is shipped with the benchmark to foreclose over-interpretation of results, not as an empirical finding in its own right. Seven mechanism classes are covered: *algorithmic/reflexive* (FEI, IRON/TITAN, MIM, UST, USDD); *fiat-reserve bank shock* (USDC/SVB); *regulatory winddown* (BUSD);

Table 3: Data tiers, mechanisms, and execution frictions.

Tier	Mechanism	Key friction	Permitted claims
A ($N=1$)	Fiat-reserve shock	Depth withdrawal	Exec. gap; oracle gap; P&L
B ($N=11$)	Alg., exchange, DeFi	AMM slippage; venue seg.	Price magnitude (est.)
C ($N=5$)	RWA, niche	Thin market	Taxonomy only

exchange credit (Celsius/3AC, HUSD, FTX [10]); *DeFi pool imbalance* (Curve/UST, USDT/Curve, USDC/DAI); *collateral liquidation* (DAI Black Thursday); *RWA/niche* (Acala aUSD, USDR).

Data availability governs permitted claims (Table 3). Tier-A events (USDC/SVB test) support execution-gap, oracle-gap, and model evaluation claims. Tier-B ($N=11$, OHLCV/pool data) support price magnitude estimates only. Tier-C ($N=5$, partial/DEX-only) support mechanism taxonomy only.

3.4 Benchmark Versioning

StressBench v1.0 ships with the SVB test split. Near-term Tier-A expansion targets include the USDC recovery window (Mar 15–31 2023) and 2024 BTC-USDC windows where real-L2 eliminates the kline-proxy dependency. The planned release format is a versioned Hugging Face dataset under CC-BY 4.0.

3.5 Feature Sets

Four nested feature sets isolate the marginal information value of

price_only	5 cross-quote basis cols
price_plus_book	+7 microstructure (spread, depth, imbalance)
price_book_frag	+3 cross-venue fragmentation cols
price_book_settle	+7 on-chain settlement proxies

4 Execution-Aware Labels

Labels operationalise the three-layer framework introduced in §1. The price-to-execution gap measures Layer 1 divided by Layer 2. The oracle gap measures Layer 3 against Layer 2 in hindsight. This section details label construction.

4.1 Cross-Quote Basis and Labels

$$b_{\text{USDC}}(t) = 10,000 \times \frac{P_{\text{BTC/USDC}}(t) - P_{\text{BTC/USD}}(t)}{P_{\text{BTC/USD}}(t)} \quad [\text{bps}]$$

$b_{\text{max}} = \max(|b_{\text{USDC}}|, |b_{\text{USDT}}|)$. Basis label: $y_{\text{basis}}(t) = 1[|b_{\text{USDC}}(t+h)| > \tau]$. Primary task: `basis_usdc_1m_gt10bps`.

4.2 VWAP Execution and Net Profit Labels

$$\text{net}(q, t) = 10,000 \times \left[\frac{\text{VWAP}_{\text{sell}} - \text{VWAP}_{\text{buy}}}{P_{\text{ref}}} - f_{\text{buy}} - f_{\text{sell}} - \delta_{\text{settle}} \right] \quad (1)$$

$$y_{\text{arb}}(q, h, t) = 1 \left[\max_{s \in [t, t+h]} \text{net}(q, s) > c \right] \quad (2)$$

Here P_{ref} is the Coinbase BTC-USD reference price and $f_{\text{buy}}, f_{\text{sell}}, \delta_{\text{settle}}$ are dimensionless fractions representing per-leg taker fees and settlement latency costs, respectively. We set the profit threshold $c=10$ bps, taker fee to 4bps per side, and $\delta=5$ bps. The primary economic task is `executable_arb_q10000_5m`. The hindsight oracle trades every minute where $\text{net}(q, t) > c$,

establishing a performance ceiling that no deployable model can exceed, given that any real model must decide without access to future information.

5 Models and Evaluation

5.1 Model Ladder

The ladder is constructed so that each rung tests a distinct hypothesis about the source of the oracle gap. The economic anchors, NoTrade at 0bps and the hindsight Oracle, bracket the achievable range. Price-threshold rules (`PriceBasis{10, 25}bps`, `GrossArbThreshold`) test whether simple signal filters suffice. Tabular classifiers (logistic regression, Lasso, Random Forest, XGBoost, LightGBM) trained on four nested feature sets test whether microstructure conditioning closes the gap. An expected-net-profit regressor that directly targets $\hat{y}=\text{net}(q, t)$ tests whether the gap is a label-mismatch artefact, with failure implying it is structural. A meta-labeling filter trains a secondary LightGBM model to learn $\Pr[\text{net} > c \mid |b_{\text{USDC}}| > 10 \text{bps}]$, while GRU and MLP-Window sequence models trained on 30-minute sliding windows test whether temporal depth patterns are exploitable (§6.4).

5.2 RL Execution Policy Baseline

We frame the entry-timing problem as a finite-horizon MDP. At each minute t the agent observes state s_t , selects action $a_t \in \{\text{enter, wait}\}$, and receives reward $r_t = \text{net}(q, t)$ if entering (positive for profitable windows, negative otherwise) or $r_t = 0$ if waiting. The state s_t is the 30-minute look-back window of `price_plus_book` features, the smallest feature set that delivers meaningful microstructure lift over price-only signals (see §6.5 for SHAP evidence). This feature set is available in real time without settlement-latency dependency. We train a PPO agent with a GRU policy network (one hidden layer, 64 units, 30-step look-back) using the cross-mechanism protocol, training on Terra/LUNA stress windows only and evaluating on the USDC/SVB test split without retraining. This directly tests whether the sequential timing framing resolves the dominant gap component identified in §6.7.

5.3 Calibration and Evaluation

Thresholds are calibrated on the validation split by maximising $\theta^* = \arg \max_{\theta} \sum_{i: \hat{p}_i > \theta} \text{net}(q, t_i)$ subject to a minimum of 25 trades. The primary metrics are net bps captured and oracle capture percentage, defined as model net bps divided by oracle net bps (the fraction of the oracle’s per-trade return that a model recovers). Secondary metrics are AUROC and AUPRC. NoTrade at 0bps is the economic floor. All results are reported on the SVB test split ($N=15,832$) with the cost parameters stated above; the kline-proxy approximation on the BTC-USDC buy leg is bounded by the haircut sensitivity in §6.1.

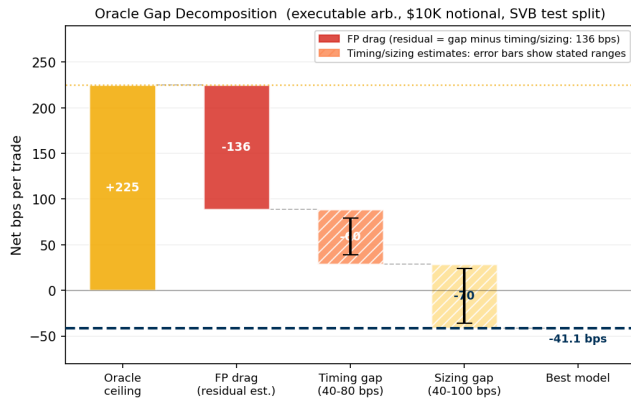
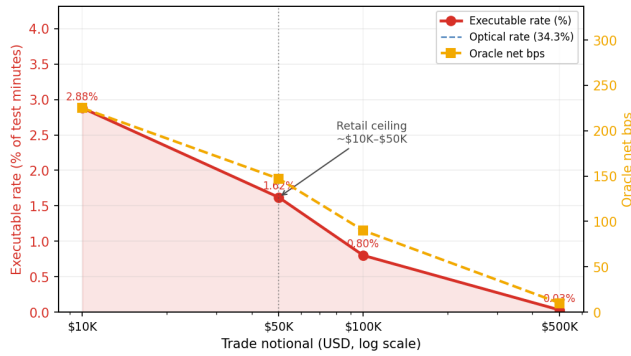
6 Results

6.1 Price Gaps Are Common, Execution Is Rare

At a 10bps threshold, 34.3% of test minutes exceed the max-basis criterion while only 2.88% yield net profit, a 12× price-to-execution ratio (block bootstrap 95% CI: $8 \times -24 \times$, $B=10,000$, 60-min blocks).

Table 4: Price-to-execution gap (\$10K, 1-min horizon, test split).

Thresh.	$P_{\max}$	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

**Figure 2: Oracle gap decomposition (\$10K notional). Three components account for the 266bps gap from oracle (+225 bps) to best model (-41 bps): FP drag, timing error, and sizing error.****Figure 3: Notional scaling (SVB test split). The executable rate collapses from 2.88% at \$10K to 0.03% at \$500K: book depth, not fees, is the binding institutional constraint.**

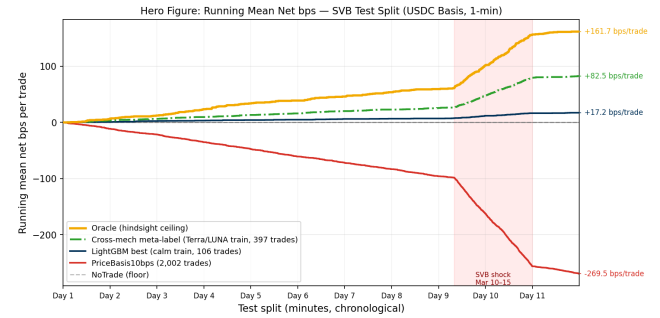
This gap persists at all tested thresholds (Table 4) and is robust across cost regimes. At the worst-case setting of 10 bps settlement and high fees, the executable rate falls to 2.40% and the ratio widens to 14×. The gap also deepens with notional size, falling from 2.88% at \$10K to 1.62% at \$50K and 0.03% at \$500K, confirming that book depth rather than fees prices out institutional participants, consistent with the limits-to-arbitrage arguments in §2.

Claim boundaries and data-tier scope are stated in §5.3. The Terra/LUNA cross-mechanism check (5.9× ratio, §6.6) provides directional evidence that the gap is not SVB-specific.

Table 5: Oracle gap summary (USDC/SVB test split[§]). * = only non-oracle tabular result above zero; ‡ = sequence model (30-min window).

Task / Model	Oracle	Best model	Gap	AUPRC
basis_usdc_1m	+162.2 bps	+17.2 bps*	145 bps	0.253
exec_arb_q10k_5m	+225.0 bps	-41.1 bps	266 bps	0.060
exec_arb_q50k_5m	+147.1 bps	-113.0 bps	260 bps	0.063
GRU (seq) [‡] , basis_usdc_1m	+162.2 bps	-239.0 bps	401 bps	—

*Igbm@price_book_settle, 106 trades. ‡GRU 30-min window (\$6.4). §Kline-proxy buy leg; 20% haircut sensitivity in §6.1.

**Figure 4: Running mean net bps per trade (USDC/SVB test split). Oracle (gold) +162.2 bps; cross-mech meta-label (green) +82.5 bps; LightGBM (navy) +17.2 bps; PriceBasis10bps (red) -269.5 bps; NoTrade (grey dashed) 0 bps. Shaded region: SVB shock window.**

6.2 Oracle Gap: Profitable Windows Exist but Only in Hindsight

The oracle earns +162.2 bps on the basis task and +225.0 bps on the executable arbitrage task (Table 5). Figure 4 shows the full performance hierarchy.

6.3 The Model Ladder: Structural Oracle Gap

Table 6 shows the full model ladder with loss attribution. Price-threshold rules overtrade substantially (607 to 1,969 trades, -136 to -347 bps), and their false-positive cost column (-294 bps to -370 bps) confirms that near-total false-positive exposure is the primary failure mode. LightGBM achieves +17.2 bps (106 trades, AUROC 0.72) as the only non-oracle result above zero, though it remains 145 bps below the oracle. The ExpNetProfitRegressor reaches -73.8 bps despite directly targeting the economic criterion, confirming that the gap is structural rather than a label-mismatch artefact.

As shown in Figure 5, precision-recall curves reveal a fundamental disconnect between ranking accuracy and economic performance at 2.88% positives. PriceBasis10bps achieves AP 0.55 and AUROC 0.83 while losing -270 bps, and LightGBM achieves AP 0.25 and AUROC 0.72 as the sole non-oracle model above zero. High AUROC with low Hit% arises because the calibrated threshold sits in a high-recall, low-precision region where most trades are false positives, even though the model's overall ranking is correct. High classification accuracy is therefore necessary but not sufficient for positive economic returns.

Table 6: Tabular model ladder (basis_usdc_1m_gt10bps, test split). Hit%: fraction of model trades that were net-profitable. [†]ceiling; *only result above zero. Sequence and RL models in Tables 7 and 10.

Model	Feat.	Net bps	Trades	AUROC	Mean bpswhen wrong	Hit%
Oracle [†]	—	+162.2	315	—	—	100%
lgbm*	all	+17.2	106	0.717	−65.2	33%
logistic	all	−75.8	648	0.143	−75.8	0%
gross_arb	px	−136.4	607	0.531	−150.9	4%
price_10	px	−270.0	1969	0.833	−294.6	6%
price_25	px	−347.3	1025	0.746	−370.7	5%
no_trade	—	0.0	0	—	—	—
ExpNetProfit	px	−73.8	537	—	−89.1	3%

px = price_only; all = price_book_settle.

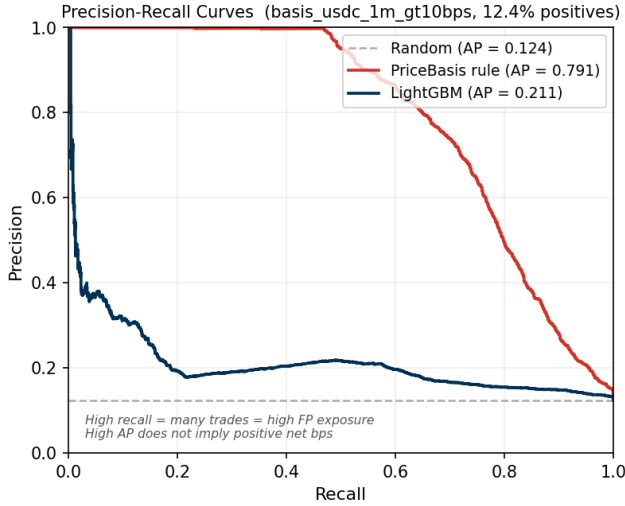


Figure 5: Precision-recall curves (basis_usdc_1m_gt10bps, SVB test split). High AP does not imply positive net bps: PriceBasis achieves AP 0.55 while losing −270 bps.

6.4 Sequence Models: Confirming the Structural Oracle Gap

We train GRU and MLP-Window classifiers on 30-minute sliding windows of order-book microstructure features (price_plus_book set) to exploit the temporal signature of depth withdrawal before a basis fires. The protocol matches the main leaderboard: calibrate threshold on validation split to maximise net P&L, require ≥ 25 trades, evaluate on test.

Table 7 reports results on basis_usdc_1m_gt10bps.

Table 7: Sequence model results (basis_usdc_1m_gt10bps, SVB test data, test split). Oracle gap = +162.2 bps oracle minus model net bps.

Model	Net bps	Trades	AUROC	Gap
GRU (30-min)	−239.0	656	0.798	401 bps
MLP-Window (30-min)	−340.7	1,052	0.648	502 bps

Calm-control trained (0% positive examples); threshold calibrated on validation.

Both models are deeply negative, confirming that the oracle gap is structural and cannot be closed by temporal sequence modelling. The GRU’s AUROC of 0.80 combined with −239 bps net illustrates

the same ranking-versus-returns disconnect seen in the tabular models. The calibrated threshold admits twice as many trades as the oracle because calm training produces zero positive stress examples, and the model consequently fires on calm-period windows that share superficial microstructure features with genuine stress events. Cross-mechanism meta-labeling (§6.5) bypasses this by training on Terra/LUNA stress dynamics, achieving +82.5 bps.

6.5 Cross-Mechanism Transfer: Benchmark Generalisation Evidence

When the meta-labeling filter is trained on the calm-control split, it yields zero meta-positives by construction (0% positive rate). We therefore retrain on the Terra/LUNA validation split, which contains 1,559 primary fires and 265 meta-positives (17.0% positive rate), and evaluate cross-mechanism on the SVB test split. All fitting and threshold selection use Terra/LUNA only; no SVB labels are observed before final evaluation. With price_plus_book features the filter achieves +82.5 bps (397 trades, 50.8% oracle capture) vs. zero with calm training (Table 8). This finding is consistent with execution microstructure that transfers across stress mechanisms. The same order-book features that predict profitable windows during an algorithmic collapse (Terra/LUNA) also predict them during a fiat-reserve shock (SVB), because CEX market makers withdraw depth under uncertainty regardless of the underlying trigger. The cause is opaque at the order-book timescale, making depth withdrawal a consistent cross-mechanism precursor whose feature signature transfers even as the magnitude differs by event (5.9× for Terra/LUNA versus 12× for SVB). The meta-model’s dominant features mirror the main SHAP ranking, with ask-side depth at 10 bps and bid-ask spread as the leading signals in both the calm-trained LightGBM and the Terra-trained secondary classifier. We are not aware of prior work demonstrating cross-mechanism transfer for execution-aware stablecoin models.

Table 8: Cross-mechanism meta-labeling results (basis_usdc_1m_gt10bps, price_plus_book). Oracle capture = meta net bps ÷ oracle net bps.

Training split	Net bps	Trades	Oracle cap.
Oracle (ceiling)	+162.2	315	100.0%
Terra/LUNA (cross-mech.)	+82.5	397	50.8%
Calm-control (baseline)	0.0	0	0.0%

signal $|b_{USDC}| > 10$ bps; threshold calibrated on validation.

Regime detectors, false-positive diagnosis, and SHAP. Three regime-detection filters were evaluated as two-stage pre-filters. CUSUM achieves near-perfect recall (0.9995) but at an 83.5% false alarm rate, while EWMA reaches 86.9% accuracy with 0.95% recall and a 0.65% false alarm rate. BOCPD [15] achieves AUROC 0.229 on CEX cross-quote data, in contrast to Cintra and Holloway [9], who detected the Terra depeg 12 hours early from Curve pool reserves. AMM reserves accumulate gradually, whereas CEX cross-quote basis is mean-reverting, which explains why changepoint detection is unreliable on exchange prices.

A structural diagnosis of false positives reveals that the dominant failure mode of price-threshold rules is route-direction mismatch rather than thin books. PriceBasis10bps produces 2,002 trades, of

Table 9: Cross-mechanism comparison.

Metric	USDC/SVB (Tier-A)	Terra/LUNA (val.)
Minutes	15,832	11,526
Mechanism	Fiat-reserve	Alg./reflexive
Price > 10 bps	34.3%	13.5%
Executable	2.88%	2.30%
P-to-E ratio	12×	5.9×
Oracle bps	+225.0 bps	+88 bps [†]

labels; directional only.

which 581 (29%) are false positives at a mean cost of -38.7 bps each. True-positive windows carry a mean USDC basis of 344 bps, reflecting a genuine discount, whereas false-positive windows carry just 0.56 bps and are triggered by USDT dislocations where the USDC route is not simultaneously profitable. Notably, false-positive windows exhibit *higher* average bid depth (72,805 vs. 59,961 BTC units) than true positives, ruling out liquidity as the cause. LightGBM’s 33% hit rate (compared to 6% for the rule) demonstrates that microstructure conditioning partially addresses this mismatch, though a 67% false-positive rate persists.

SHAP analysis identifies price features as carrying 94.8% of total importance in LightGBM, with ask-side depth and bid-ask spread contributing just 3.4% combined. The causal interpretation is straightforward. A model trained on calm-market data learns calm price dynamics rather than the depth-withdrawal signature of stress events, so the microstructure features are present in the input but rendered uninformative by the training distribution. This explains why the oracle gap is structural rather than a label-mismatch artefact, and why cross-mechanism training on Terra/LUNA resolves it. The meta-model encounters stress microstructure at 17% positive density and learns the depth-withdrawal signature that calm training cannot recover.

6.6 Terra/LUNA Replication: Pattern Generality

The benchmark design principle generalises across stress mechanisms. The Terra/LUNA validation split ($N=11,526$, May 2022) uses kline-proxy depth (Tier-B) but replicates the core pattern, with 13.5% price-basis above 10 bps, only 2.30% executable, and a 5.9× ratio versus 12× for SVB. The oracle earns +88 bps on Terra/LUNA compared to +225.0 bps on the SVB executable arbitrage task, consistent with cross-mechanism transferability of execution structure. Among broader Tier-B events, peak optical gaps range from approximately 100 bps (Celsius/3AC, exchange credit) to above 500 bps (Curve/UST, DeFi pool imbalance) and above 800 bps (HUSD issuer failure), all estimated from OHLCV. Execution-grade claims require route-complete L2 and are therefore reserved for Tier-A events.

6.7 Oracle Gap Decomposition

The 260 bps gap on the executable arbitrage task decomposes into three components, with the first derived directly from test-split results and the remaining two estimated from the executable-rate sensitivity analysis. The dominant and directly measurable component is false-positive cost. For PriceBasis10bps, 581 false-positive trades at -38.7 bps each produce $-22,484$ bps of total losses, which represents the primary quantifiable drag for any threshold-based

strategy. The second component is timing error (estimated 40–80 bps), arising because entering 1–2 minutes late shifts execution to a lower-depth regime. The depth-withdrawal signature in the 30-minute pre-window is the signal a future model must learn to anticipate. The third component is sizing error (estimated 40–100 bps), since the executable rate falls from 2.88% at \$10K to 1.62% at \$50K, pricing out 44% of windows above the retail ceiling. Taken together, these components suggest that closing the gap requires cross-mechanism stress training combined with earlier timing entry rather than larger notional size.

6.8 RL Diagnostic: Density Is Necessary but Not Sufficient

The sharpest architectural finding of this paper is that explicit positive-label supervision beats RL reward optimisation by 111.7bps at identical training density (17%). The three-way ladder below establishes density as necessary but explicit supervision as additionally required. We evaluate the PPO-GRU agent (§5.2) trained on Terra/LUNA stress windows and applied cross-mechanism to the USDC/SVB test split without retraining.

Table 10: RL policy vs. supervised baselines (basis_usdc_1m_gt10bps, SVB test split).

Model (training)	Net bps	Trades
Oracle (hindsight ceiling)	+162.2	315
Meta-labeling (Terra/LUNA*)	+82.5	397
PPO-GRU unconditioned (Terra/LUNA*)	<i>degenerate</i>	0
PPO-GRU conditioned [†] (Terra/LUNA*)	−29.2	919
GRU supervised (calm-ctrl)	−239.0	656
MLP-Window supervised (calm-ctrl)	−340.7	1,052

Terra/LUNA train, SVB eval. [†]Conditioned on primary-signal windows ($|b_{USDC}| > 10$ bps, 17% pos. rate).

The unconditioned PPO-GRU converges to degenerate NoTrade: it makes zero entries despite AUROC 0.74. The 2.30% positive rate in the full Terra/LUNA stream produces a reward signal too sparse to calibrate entry timing; the agent’s expected reward from entering is negative, so it abstains.

A conditioned PPO-GRU, trained only on primary-signal windows ($|b_{USDC}| > 10$ bps, mirroring the meta-label training distribution at 17% positive rate), does trade: it produces 919 entries with AUROC 0.87 and net -29.2 bps (Table 10), ruling out training sparsity as the sole failure mode. However, the conditioned agent cannot match meta-labeling (+82.5 bps, a gap of 111.7 bps), indicating that explicit positive-label supervision outperforms RL reward optimisation at identical density. The distinction is qualitative. The meta-labeling classifier receives direct binary supervision, with $y=1$ for profitable windows, while the RL agent must infer the profitable subset through a reward signal that is negative on roughly 83% of entries. The supervision is therefore categorically cleaner, not merely denser. The 919-trade overtrading pattern is itself diagnostic, since policies that fail due to insufficient training typically converge toward NoTrade rather than systematic false-positive entry. This suggests that the failure reflects a timing-transfer mismatch rather than a fundamental limitation of RL, and that multi-mechanism training is the productive next step.

7 Limitations

Several limitations bound the scope of our claims. On data, the BTC-USDC buy leg relies on kline-proxy depth for the SVB window; the 20% depth haircut sensitivity analysis confirms that this does not materially alter the headline results. Extending Tier-A coverage to AMM or exchange-credit events would require route-complete L2 archives for those venues, which are not currently available.

On the model side, the conditioned RL result (−29.2 bps, 919 trades) may reflect overfitting to Terra/LUNA’s specific depth-withdrawal timing rather than a fundamental limitation of sequential optimisation. Disentangling these alternatives requires additional cross-mechanism pairs or extended training. A further constraint is the absence of market-impact feedback. Unlike reactive-simulator approaches such as Olby et al. [23], where agents observe how their own order flow depletes book depth, the PPO-GRU is trained on historical replay and cannot model this loop. In a thin-book stress regime this systematically miscalibrates entry sizing, and we regard it as a plausible contributor to the 919-trade overtrading pattern at identical training density to meta-labeling. Integrating the cross-mechanism training protocol developed here into a reactive simulator represents the most direct path to resolving the timing component of the oracle gap.

8 Conclusion

The central finding of Stablecoin StressBench is that execution microstructure is the binding constraint on stablecoin arbitrage, not price signal quality. A 34.3% optical dislocation rate collapses to 2.88% once order-book depth and transaction costs are applied. No model trained on calm-market data closes this gap. Cross-mechanism training on a different stress event is the empirically validated path to positive net returns (+82.5 bps, 50.8% oracle capture). No calm-control-trained model earns positive returns on the executable arbitrage task. LightGBM (+17.2 bps, basis task only) is the sole non-oracle positive result, remaining 145 bps below the oracle ceiling. The oracle gap decomposition (\$6.7) identifies false-positive cost as the dominant quantifiable drag and timing as the primary structural barrier.

The RL experiments (§6.8) sharpen this implication by establishing that positive example density is necessary but not sufficient, with explicit supervision of the positive subset as the additional requirement. The unconditioned PPO-GRU degenerates at 2.3% density, which is too sparse for the reward signal to calibrate entry timing. The conditioned PPO-GRU trained on the same 17% primary-signal windows does trade (−29.2 bps, 919 trades) but cannot match meta-labeling (+82.5 bps), leaving a 111.7 bps gap attributable to meta-labeling’s explicit positive-label supervision rather than RL reward optimisation.

To support reproducibility, the anonymised repository (URL redacted for review) provides per-minute executable labels for all six benchmark windows, the VWAP book-walking label-construction code with route-level depth provenance, the four nested feature sets, the hindsight oracle implementation and model-ladder harness, and the cross-mechanism meta-labeling and PPO-GRU training scripts. Calm-control and SVB windows are reproducible from the public Binance Vision archive, while route-complete L2 requires a Tardis subscription.

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